Marker cube brief

Nov 3, 2026 · Ben Zheng, MS2, for Emily Simpson

What the project is

We are building a "completeness coach" for ultrasound training: a small printed marker cube rides on the probe, a camera watches the cube, the ultrasound video is captured from the machine's HDMI output, and software builds a 3-D map of which parts of the target volume the student has actually imaged. The coach then names what was missed ("left lobe, lower pole") and gives a completeness score. It does not read the images for pathology; it only answers "did you look everywhere?"

It is a student research project under Dr. Patrick Brooks as faculty mentor. We are submitting it for the AWSOM MS2 Student Research Seed Grant (due December 2) as a two-phase pilot: phantom testing first, then 16 student scanners doing thyroid sweeps on volunteers, scored against two blinded faculty raters.

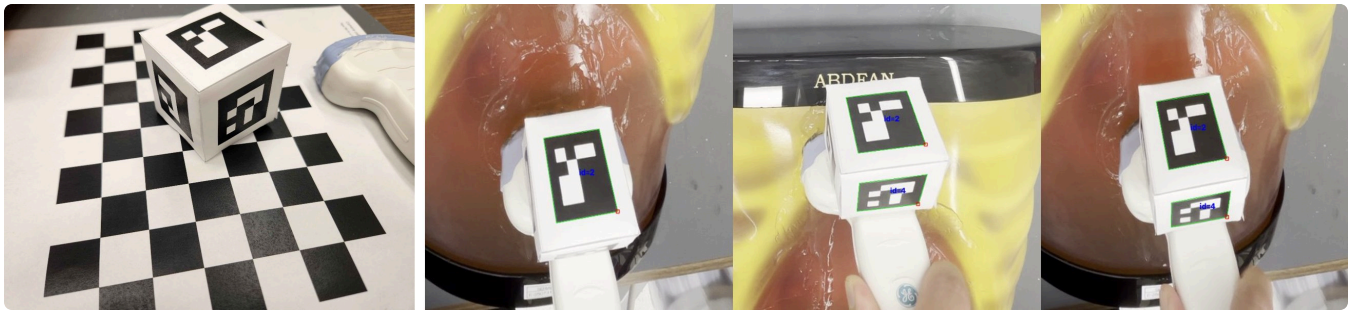
Team: Ben Zheng, Sally Lin and Ashleigh Lacey (MS2s); Vahdeta Suljic (engineering partner, builds and runs the rig and software); Dr. Kevin Kunkler provides the GE Venue Go and simulation lab time.

What we have done so far

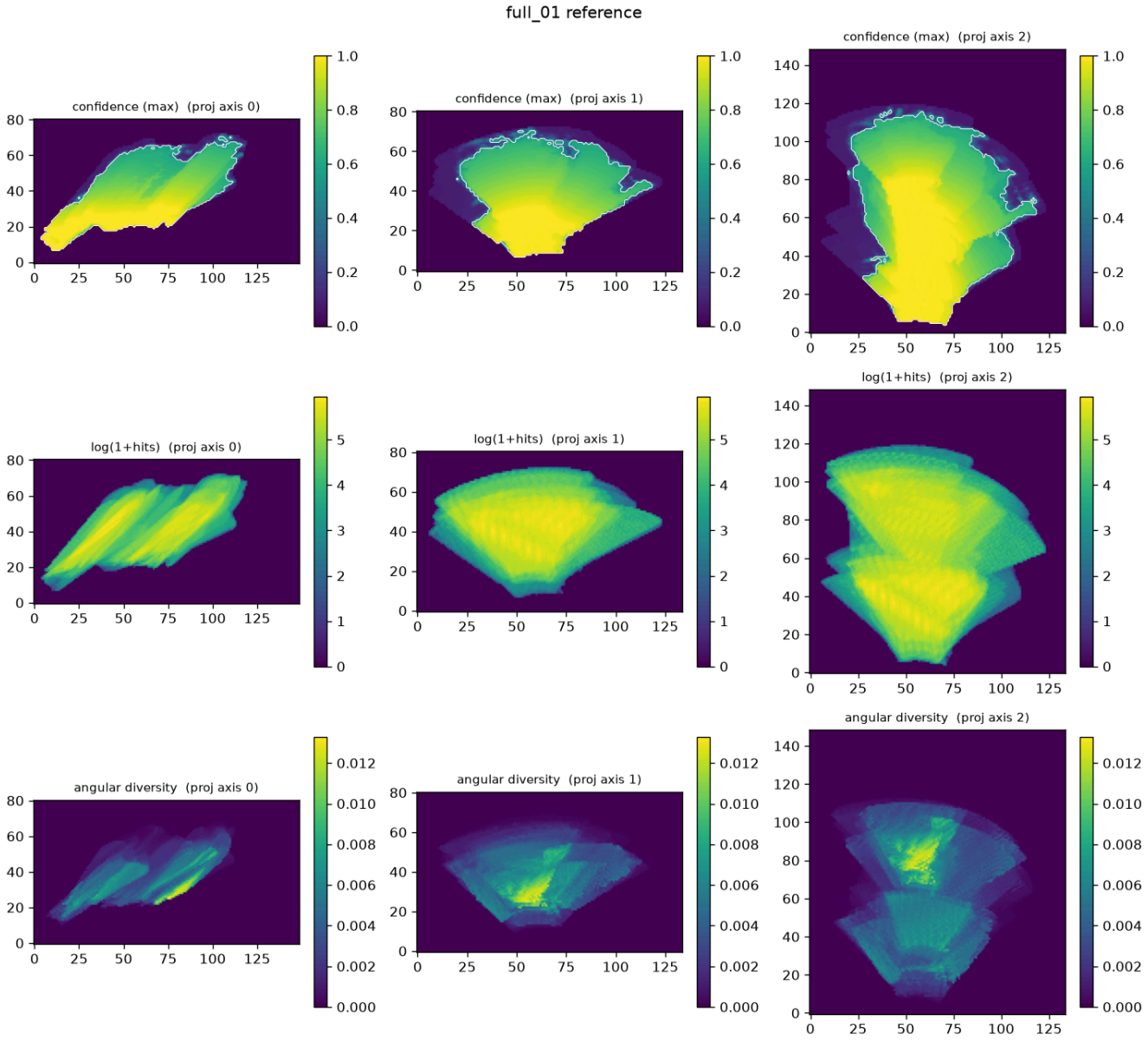
The software exists and has been through two lab sessions on the GE Venue Go in the simulation lab with the abdominal phantom.

- **Session 1 (October 22).** Everything ran end to end, but the tracking was poor: the cube was tracked in only about a quarter of frames. The causes were all rig-related: markers printed without a white border, the hand covering the cube, and the camera too far away.
- **Session 2 (October 30).** With a white paper cube, bordered markers and the camera about 30 cm away, the cube was tracked in 87 to 97 percent of frames during sweeps. The probe held still moves less than half a millimetre in the track, a known slide along a chessboard was measured to within 2.3 mm on average, and we got our first real coverage maps. Two of us did the same sweep and the maps overlapped by about 70 percent.

The current cube is a hand-made paper cube, about 57 mm on a side, with 35 mm ArUco markers glued on, taped to the back of the probe. It works, but it is fragile, it sits slightly differently every time we attach it, and we have to rebuild it for each session. That is why we would like a printed one.



Left: the paper cube we use today, on the chessboard we use as a ruler. Right: what the tracking camera sees, with both marker faces detected.

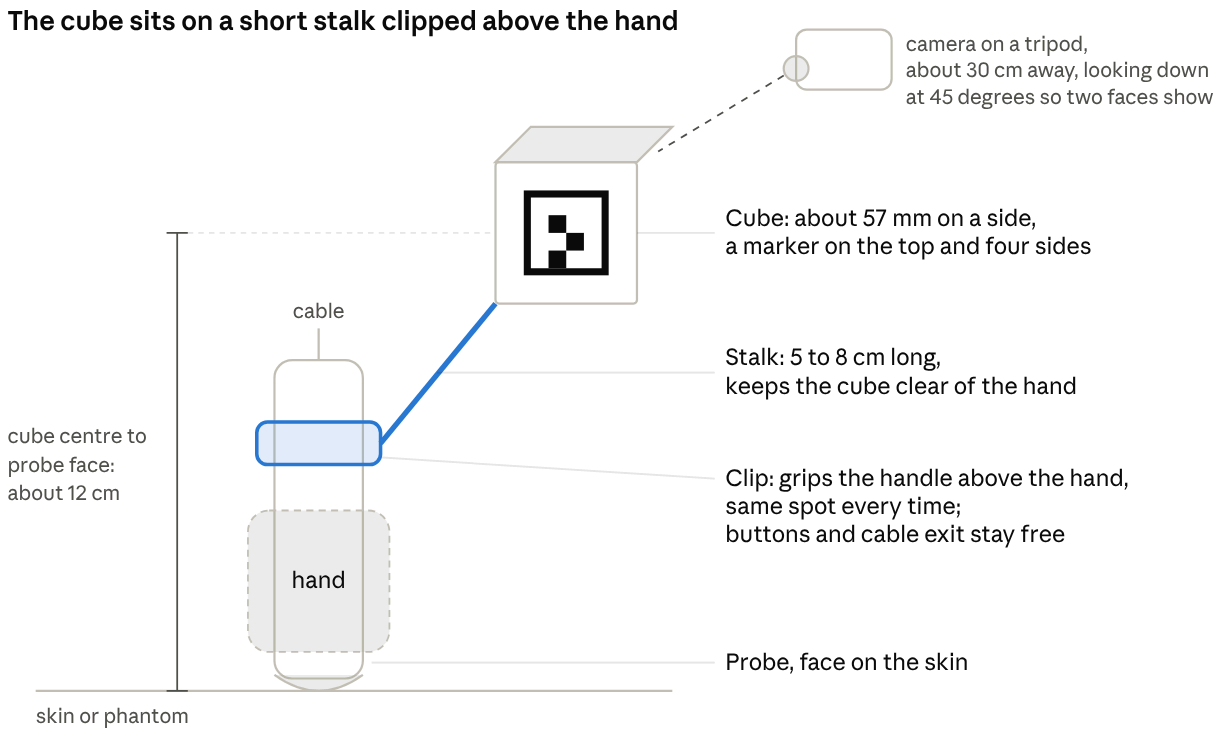


Coverage maps from one full sweep on the phantom. Each column is a different viewing direction. Top row: how confidently each part was imaged (yellow is fully covered; the white line is the edge of the covered volume). Middle row: how many frames hit each spot. Bottom row: how many angles it was seen from. This is what the cube's tracking turns into.

What we are asking for: a 3D-printed marker cube

We do not have a 3D model yet. What we have is the paper version, the dimensions below, and a clear idea of what matters. We would be glad to sketch it with you, or send a rough STL if that is easier to start from.

Part	What we need	Why it matters
Cube body	About 57 mm on a side (55 to 60 mm works; we enter the final size in the software). Hollow and as light as practical.	Matches our marker sheets. Larger is more accurate but heavier on the probe.
Marker faces	Five faces (top and four sides), each with a shallow recess about 52 x 52 mm and 0.5 mm deep to seat a paper marker sticker flush. Faces flat and matte.	Each sticker is a 35 mm marker plus an 8 mm white border. Detection needs a flat, crisp square with white around it.
Colour and finish	White or light grey, matte.	Contrast for the markers and no glare under lab lights.
Face labels	A small embossed number (1 to 5) on each face and an arrow for up.	So the markers go on the same way every time and the software knows which face is which.
Mount	A short stalk, 5 to 8 cm, from the cube's bottom face down to a clip or saddle that grips the probe handle behind the hand. Rigid, no play, seats in the same position every time; a snap fit or a two-piece clamp with one screw rather than a friction sleeve. Leaves the probe buttons and cable exit free.	The map is only as good as the fixed geometry between cube and probe; any wobble becomes measurement error. The stalk keeps fingers off the marker faces.
Probes	GE Venue Go. We have used the C1-5 curvilinear probe so far; the thyroid study will use the L4-12t-RS linear probe. One complete print per probe (clip, stalk and cube as a single piece) is fine; a swappable clip only if that is easier for you.	Every joint is a place the cube can shift, so fewer parts is better; we calibrate each print once.
Quantity	Two cubes with clips, so we have a spare.	A session is lost if the only cube breaks.



side view of the probe with clip, stalk and cube

Side view, not to scale. Today the cube is taped straight onto the handle; the clip and stalk replace the tape so the cube sits clear of the hand and in the same place every time. The 12 cm is what we measured on the paper rig and can change; we calibrate it once per cube.

Nice to have, not required: if the stalk can tilt the cube so that two faces point toward a camera placed at about 45 degrees, tracking gets more accurate. We can live without it.

Timeline

- **Now to December 2:** finishing the seed grant application. The cube is already in the proposed budget as a small supplies item.
- **Next lab session:** not yet scheduled, probably late November or December. Having a first print to try then would be ideal, and we are happy to test a rough version and send feedback for a second pass.
- **If funded, December 16 onward:** phantom testing for the first two months, then volunteer scanning in the spring and a poster at AWSOM Research Day. The cube would be used throughout, so a version we are happy with by mid-December would carry the whole study. There is no hard deadline on the cube; whatever timing works for you, we will plan the next session around it.